

Deepak Titus Edward

Staff Software Engineer | Generative AI, RAG & Agentic AI Systems

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SUMMARY

Staff Software Engineer with 18+ years building large-scale distributed systems, real-time data pipelines and cloud-native platforms, now leading production Generative AI. Architected and shipped an AWS Bedrock Retrieval-Augmented Generation (RAG) assistant with an LLM-as-a-judge evaluation framework that raised answer correctness from 0.17 to 0.83, and designed a 5-agent, LangGraph-based next-best-action system on Claude with a three-layer safety stack (LLM critic, deterministic guardrails, human-in-the-loop). Deep experience in Customer Data Platforms (CDP), streaming data engineering on Databricks and Kafka, and AWS, at 100M+ device scale. Known for end-to-end ownership, cross-functional leadership and measurable production outcomes.

SKILLS

Generative AI & Agents: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI agents, multi-agent systems, LangGraph, AWS Bedrock (Agents, Knowledge Bases, Guardrails), Anthropic Claude, Amazon Nova, prompt engineering, LLM evaluation (LLM-as-a-judge), structured outputs, semantic caching, embeddings, reranking, human-in-the-loop, Responsible AI

Vector Search & ML: OpenSearch Serverless, PostgreSQL pgvector, Cohere Rerank, Titan Embeddings, Databricks Mosaic AI Model Serving, Databricks Genie, scikit-learn, XGBoost, model evaluation (F1, ROC-AUC)

Data Engineering: Databricks, Apache Spark Structured Streaming, Delta Lake, Unity Catalog, Apache Kafka, Avro, medallion architecture, SQL, ETL/ELT, data modeling, data quality, identity resolution

Cloud & Platform: AWS (Lambda, Step Functions, EventBridge, SQS, DynamoDB, ECS Fargate, S3, CloudWatch, IAM), Kubernetes, Kong API Gateway, Terraform, GitHub Actions, CI/CD, observability, SLOs

MarTech & CDP: Adobe Experience Platform (RT-CDP), Adobe Analytics, Tealium, Hightouch, Braze, LiveRamp, Akamai, audience activation, GDPR/privacy engineering

Languages: Python, SQL, Java, JavaScript

EXPERIENCE

Staff Software Engineer

McAfee | Santa Clara, CA | Jul 2021 – Present

Lead MarTech data engineering and Generative AI for the consumer business on AWS, Adobe Experience Platform and Databricks, at 100M+ device scale.

- Built a 5-agent next-best-action system on Claude that reads shopper clickstream (landing page, cart, checkout) and decides for each cohort when to reach out, which offer to send and on which channel (email, SMS, onsite) to recover abandoned carts and drive renewals.
- Coded the agent workflow in Python with LangGraph: timing, offer and channel agents run in parallel, an orchestrator merges them into one typed decision, and a critic agent rejects weak or unsafe decisions; durable checkpointing lets runs resume after a crash.
- Implemented a three-layer safety stack: the LLM critic, deterministic Python guardrails (20% discount ceiling, SMS consent, quiet hours) and human review when confidence is below 0.6 or the cart is \$150+.
- Deployed the agent system Databricks-native (Unity Catalog, Mosaic AI Model Serving, Delta tables, Databricks Asset Bundles) with Braze activation, idempotent decisions and a Streamlit review console.
- Architected and shipped a production RAG assistant on AWS Bedrock (Agents, Knowledge Bases, Guardrails) over a multi-locale knowledge base, with least-privilege IAM and dual vector stores (OpenSearch, pgvector).
- Built an LLM-as-a-judge evaluation framework over ~500-prompt datasets across 5 iterations, raising answer correctness from 0.17 to 0.83 with citation coverage of 0.88 and harmfulness of 0.00.
- Engineered a verified semantic cache (direct answer above 80% similarity, few-shot at 60–80%, full agent below 60%) with reranking to reduce latency, cost and hallucinations; root-caused and fixed a production hallucination in knowledge-base source data.

- Defined production SLOs (99.9% availability, P95 latency under 2 s, error rate under 0.1%) with CloudWatch alarms, per-model token-cost dashboards and incident runbooks.
- Designed a first-party streaming event pipeline handling 15M events/day (500K/min peak): a dependency-free JavaScript SDK, Kong, Kafka with separate PII and non-PII Avro topics, and a Databricks medallion with idempotent Spark Structured Streaming.
- Led a first-party-domain tag migration behind an Akamai reverse proxy covering 2–5B hits/year, coordinating 6 teams (MarTech, InfoSec, PKI, Network, Akamai, Tealium) with a validation-gated runbook.
- Modeled customer data at 100M+ device scale, led a device-schema migration, found a critical identity-stitching defect, and built activation audiences of 60M+ profiles across 8+ AdTech destinations.
- Built an XGBoost dynamic-pricing classifier; implemented GDPR pipelines and 30+ CloudWatch dashboards with anomaly detection; automated profile cleanup, eliminating 2.5 hours/day of manual work.
- Authored the company's Senior AI Engineer hiring framework and 5-round interview loop with a weighted scorecard; interviewed ~10 candidates across two teams. Recognized for Outstanding Performance and Excellence in Execution.

Senior Technical Architect

HCL Technologies | San Francisco Bay Area, CA | Jul 2020 – Jul 2021

- Architected business-critical Java microservices (Spring Boot, Spring Batch) on Kubernetes and scalable data-ingestion tooling; led architecture reviews with product stakeholders.

Staff Software Engineer, Lead Monetization

realtor.com | San Francisco Bay Area, CA | Aug 2018 – Jun 2020

- Built large-scale distributed REST microservices on AWS (Spring Boot); migrated AWS Lambda services to an ECS Spring Batch application for scalability.
- Built Splunk and New Relic observability; mentored engineers and set engineering best practices.

Technology Lead

Infosys | San Francisco Bay Area, CA | Apr 2015 – Jul 2018

- Lead Java backend engineer for Apple's product-manufacturing team; designed multithreaded REST APIs with test-driven development and led front-end and back-end developers.

Lead Software Engineer / Senior Software Engineer

ITC Infotech | San Francisco Bay Area, CA and Bengaluru, India | Jul 2007 – Apr 2015

- Designed REST APIs on Google Cloud Platform for Motorola Mobility; core member of the Big Data Center of Excellence (Hadoop: Cloudera, Hortonworks).
- Module lead for a UK online-banking portal and the Starwood Hotels reservation system.

EDUCATION

Post Graduate Program in Artificial Intelligence and Machine Learning, The University of Texas at Austin, McCombs School of Business, 2024

Master of Computer Applications (MCA), Computer Science, Dr. A.P.J. Abdul Kalam Technical University, 2007

Bachelor of Science, Mathematics, Chemistry and Physics, Kanpur University, 2004

CERTIFICATIONS

- AWS Certified AI Practitioner, Amazon Web Services (2025)
- Databricks Certified Generative AI Engineer Associate, Databricks
- Generative AI for Practitioners; Introduction to Generative AI; Machine Learning Foundations; AI on Cloud